

MOLDE ARO, Norway

WMO: 012170

Lat: 62.745N

Lon: 7.263E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	20.0	8.0	8.3	23.1	11.3	9.8	24.2	33.4	38.9	28.4	40.1	9.6	30	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	75.2	62.7	71.6	60.9	68.4	59.5	64.3	71.8	62.3	69.1	60.5	66.9	6.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	80.0	68.8	59.1	74.8	66.0	57.4	70.4	63.9	29.5	72.2	28.0	69.2	26.8	67.1	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.7	23.9	20.3	14.0	82.6	3.2	3.9	11.7	85.4	9.8	87.6	8.0	89.8	5.7	92.6		
(5)			13.1	67.9	3.2	3.0	10.8	70.0	8.9	71.7	7.1	73.4	4.8	75.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	33.1	35.2	37.3	42.3	49.6	55.3	59.9	58.6	53.2	45.3	38.2	36.2
	DBStd	10.94	6.17	6.19	5.55	5.67	6.91	5.21	5.45	4.55	4.77	5.52	6.65	6.48
	HDD50	2652	523	413	394	242	93	11	1	2	23	165	357	428
	HDD65	7179	988	833	859	682	478	295	177	204	353	612	805	893
	CDD50	976	0	0	0	10	81	169	309	268	120	17	2	0
	CDD65	31	0	0	0	0	1	4	21	5	0	0	0	0
	CDH74	153	0	0	0	0	7	22	106	18	0	0	0	0
	CDH80	19	0	0	0	0	0	2	16	1	0	0	0	0
Wind	WSAvg	8.8	9.9	9.4	10.0	8.5	7.8	8.0	7.1	7.2	8.8	8.7	9.6	11.2
Precipitation	PrecAvg	65.5	6.2	6.1	6.3	3.3	3.3	4.2	3.9	5.0	6.7	6.7	6.3	7.3
	PrecMax	84.1	14.3	14.4	18.2	8.9	6.6	9.9	7.6	11.9	17.6	12.4	15.2	15.3
	PrecMin	43.1	0.0	0.0	1.3	0.3	1.0	1.0	1.0	0.7	1.7	0.8	0.1	1.0
	PrecStd	10.1	4.2	3.6	4.3	2.2	1.5	2.3	1.8	2.6	4.1	2.9	4.4	3.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	51.3	53.5	67.6	74.8	77.1	82.4	76.1	69.0	63.4	55.4	52.1
		MCWB	42.1	43.2	43.3	52.8	59.9	62.5	66.5	63.1	58.3	53.4	48.8	44.7
	2%	DB	45.5	48.1	49.9	60.4	69.9	71.7	76.9	71.9	65.5	59.3	51.9	49.2
		MCWB	41.3	41.7	42.9	49.7	57.9	60.0	64.5	60.7	56.5	51.3	46.7	43.6
	5%	DB	43.3	45.9	47.7	55.3	64.8	67.5	72.7	68.9	62.9	55.5	49.4	47.1
		MCWB	39.6	40.5	42.5	47.5	55.2	57.6	63.1	60.4	55.9	49.5	45.4	42.3
	10%	DB	41.1	43.7	45.7	51.6	60.5	64.1	68.6	66.3	60.6	52.8	46.7	45.0
		MCWB	37.4	39.0	40.9	44.8	52.7	55.8	60.8	59.1	54.9	48.2	43.0	40.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	45.5	46.5	53.8	61.1	63.0	68.8	64.6	60.9	54.8	50.3	47.3
		MCDB	47.1	48.4	49.2	64.7	71.8	72.8	78.6	71.6	66.3	59.9	53.5	49.7
	2%	WB	42.3	43.1	44.7	50.7	58.5	60.5	66.0	62.8	58.7	52.4	48.4	45.2
		MCDB	44.4	46.2	48.1	59.2	67.7	69.7	74.9	69.1	63.1	57.7	51.4	48.0
	5%	WB	40.0	41.3	42.9	48.3	55.9	58.5	63.7	61.1	57.0	50.4	45.6	43.1
		MCDB	42.6	45.1	47.4	54.4	63.7	66.2	70.9	66.8	61.2	54.9	48.5	46.1
	10%	WB	38.1	39.3	40.9	45.7	53.7	56.6	61.6	59.8	55.5	48.4	43.5	41.3
		MCDB	40.5	43.2	44.7	50.2	60.1	62.7	68.1	65.2	59.7	52.1	46.2	44.9
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.9	11.2	13.4	14.1	12.5	12.2	12.4	11.3	10.7	8.6	8.4
		MCDBR	10.1	10.8	12.8	20.5	21.0	18.9	18.4	16.9	14.2	13.4	11.2	11.1
		MCWBR	7.5	7.4	8.5	11.8	11.3	10.0	9.6	9.0	8.6	8.9	8.5	8.1
	5% WB	MCDBR	9.3	9.8	11.4	18.4	19.4	16.7	17.1	15.0	12.6	11.8	10.2	10.7
		MCWBR	7.3	7.4	8.2	10.9	10.8	9.5	9.3	8.7	8.4	8.2	8.4	8.5
Clear-Sky Solar Irradiance	taub	0.241	0.296	0.324	0.347	0.347	0.350	0.360	0.351	0.331	0.307	0.249	0.358	
	taud	2.082	2.265	2.317	2.355	2.414	2.468	2.479	2.502	2.539	2.452	2.147	2.597	
	Ebn at Noon	129	207	247	266	276	277	271	263	247	206	125	75	
	Edh at Noon	10	19	26	31	32	31	30	27	22	16	9	6	
All-Sky Solar Radiation	RadAvg	46	193	540	1052	1446	1548	1448	1112	650	293	75	26	
	RadStd	9	41	87	114	88	155	122	107	90	46	19	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		+0.90	+1.22	N/A	N/A	N/A	N/A	-270	-394	+113	N/A

Nomenclature: See separate page